

PARTS LIST

Semiconductors:

- | | |
|------------|--------------------------|
| IC1 | - LM555 timer |
| D1-D4 | - 1N4004 rectifier diode |
| D5-D6 | - 1N4148 signal diode |
| IRF1 | - 60NF06 MOSFET |
| LED1, LED2 | - 5mm LED |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|-------|------------------------|
| R1-R3 | - 1-kilo-ohm |
| R4 | - 100-ohm |
| R5 | - 100-kilo-ohm |
| VR1 | - 47-kilo-ohm potmeter |

Capacitors:

- | | |
|--------|---------------------------------|
| C1, C2 | - 100 μ F, 25V electrolytic |
| C3, C5 | - 100nF ceramic disk |
| C4 | - 10nF ceramic disk |

Miscellaneous:

- RL1 - 12V, 1C/O relay
- CON1 - 7-pin connector
- CON2 - 2-pin connector
- DRL - Daylight running lights

Set Off mode as per your choice, or according to the relevant law of the land.

The default Set Off mode of the DRL is given below:

IGN+ (ignition): CN→DRL: ON

HL+/PL+/PB+ (headlight/park light/hand
brake): ON→DRL: OFF

The circuit is a simple pulse-width modulator (PWM) built around the ubiquitous 555 timer. User-controllable PWM output from IC1 is used to switch on the DRLs through MOSFET 60NF06 (as MOSFET on DRL ground is connected to circuit ground).

Here, 555 is configured as astable and, hence, it is possible to have completely-independent control of charge and discharge times of the timing capacitor by using two external diodes (D5 and D6). The 12V 1C/O electromagnetic relay in the circuit is used to enable/disable the DRL controller circuitry, as per status



Fig. 3: Photograph of the DRL

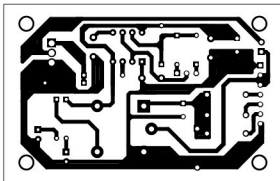


Fig. 4: Actual-size PCB of the DRL

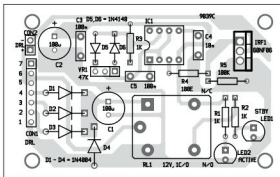


Fig. 5: Component layout of the PCB

of the headlight/parking light/hand brake. LED1 indicates standby and LED2 indicates the active modes of the DRL controller.

Note. Driving the MOSFET from a noisy line calls for a small series gate resistor close to the MOSFET. Using a low-value 100-ohm resistor

(R4) between the MOSFET driver and MOSFET gate terminal dampens down any ringing oscillations caused by lead inductance and gate capacitance, which can otherwise exceed the maximum voltage allowed on the gate terminal. Also, using pull-down 100k resistor (R5) from the gate to the source of the MOSFET is a good practice.

Construction and testing

An actual-size, single-side PCB for the DRL controller circuit is shown in Fig. 4 and its component layout in Fig. 5. Enclose the circuit in a suitable small box with connectors CON1 and CON2 on the front side to connect the seven control signals and the DRL.

After assembling the circuit, refer to the wiring guide table before connecting these to the PCB board.

Panel-mount the input and output interface, as required. ●



T.K. Hareendran is an electronics hobbyist, freelance technical writer and circuit designer

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